

Ex. No. : 6

SELECT with various clause – BETWEEN, IN, Aggregate function

1. List the type of the courses “Statistics” and “Programming”

```
mysql> select ctype from course where cdesc in('statistics','python');
+-----+
| ctype |
+-----+
| practical |
+-----+
1 row in set (0.00 sec)

mysql>
```

2. The instructor wants to know the CourseNos whose scores are in the range 50 to 80

```
mysql> select cno,score from mark where score between 50 and 80;
+-----+-----+
| cno | score |
+-----+-----+
| c004 | 75 |
| c002 | 60 |
| c003 | 70 |
| c002 | 50 |
+-----+-----+
4 rows in set (0.01 sec)
```

AGGREGATE

1. Find the average mark of “C002”.

```
mysql> select avg(score) as average_mark from mark where cno='c002';
+-----+
| average_mark |
+-----+
| 68.3333 |
+-----+
```

2. List the maximum, minimum mark for “C002”

```
mysql> select max(score) as maximum,min(score) as minimum from mark where cno='c002';
+-----+-----+
| maximum | minimum |
+-----+-----+
| 95 | 50 |
+-----+-----+
1 row in set (0.00 sec)
```

3. List the maximum, minimum, average mark for each course

```
mysql> select cno,max(score) as maximum, min(score) as minimum,avg(score) as
average from mark group by cno;
```

cno	maximum	minimum	average
c001	85	85	85.0000
c002	95	50	68.3333
c003	70	70	70.0000
c004	75	75	75.0000

4.List the name of the courses and average mark of each course.

```
mysql> select course.cdsc,avg(mark.score)as average_mark from course join m
ark on course.cno=mark.cno group by course.cdsc;
```

cdsc	average_mark
python	85.0000
dtabase	68.3333
statistics	70.0000
science	75.0000

5.Calculate the sum of all the scores.

```
mysql> select sum(score) as total_score from mark;
```

total_score
435

1 row in set (0.00 sec)

6. How many students are registered for each course? Display the course description and the number of students registered in each course.

```
mysql> select course.cdsc,count(student.regno) as no_of_students from cours
e join student on course.cno=student.cno group by course.cdsc;
```

cdsc	no_of_students
python	2
dtabase	2
statistics	1
science	1

4 rows in set (0.02 sec)

7. How many courses did each student register for?

```
mysql> select regno,count(cno)As total_courses from mark group by regno;
```

regno	total_courses
192572134	1
192572135	1
192572136	1
192572137	1
192572138	2

```
5 rows in set (0.00 sec)
```